

Group I, Row V, part (b): the UV log in transverse coordinates

Replacing $\log \frac{\mu^2}{k^2}$ by $\log(\mu^2(y-x)^2)$: the coordinate-space counterterm, done in $d = 2 - 2\epsilon$, then the p^+ integral — every step, each one checked numerically

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi, \quad q = \bar{\xi}k, \quad D = (r - s)^2 + \frac{\xi}{\bar{\xi}}s^2, \quad \lambda = \frac{\Lambda}{k^+}, \quad L = \log \frac{k^+}{\Lambda}, \quad \mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}, \quad C_{UV}(\xi) = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}, \quad \mathcal{W} = \frac{P \cdot s}{P^2 s^2}, \quad \mathcal{T} = \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2}$$

The answer.

$$\frac{d^3 N_b}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \left\{ \mathcal{W} \left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{MS}^2 (y-x)^2 e^{2\gamma_E}}{4} \right] + \frac{1}{2} \mathcal{W} L^2 + \frac{\tilde{\beta}_0 + \tilde{\beta}_1}{\pi} L + \frac{\tilde{\mathcal{F}}_b}{\pi} \right\} \mathcal{U}_b$$

- The pole and its coefficient $2L - \frac{11}{6}$ are the same as before. Only the scale in the log changes, from $\frac{1}{k^2}$ to $\frac{(y-x)^2 e^{2\gamma_E}}{4}$.
- The log vanishes at $\mu_{MS}^2 = \frac{c_0^2}{(y-x)^2}$ with $c_0 = 2e^{-\gamma_E}$, the usual coordinate-space scale.
- $\tilde{\beta}_{0,1}$ and $\tilde{\mathcal{F}}_b$ are finite and free of ϵ and Λ (§7). The total is identical to the momentum-space result of `rowV_full_derivation.pdf` (§8); only the split between the UV term and the remainder differs.

0. Where we start, and why the log came out in k

After the w integration, part (b) of the .tex is (see `rowV_full_derivation.pdf`, §§0–2)

$$\frac{d^3 N_b}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \int_{\lambda}^{1-\lambda} d\xi \int d^2 r e^{iq \cdot r} \frac{T(r)}{\xi D} \mathcal{U}_b, \quad \mathcal{U}_b \text{ independent of } z.$$

As $r \rightarrow 0$ we have $D \rightarrow s^2 + \frac{\xi}{\bar{\xi}}s^2 = \frac{s^2}{\bar{\xi}}$, so $\bar{\xi}D \rightarrow s^2$ and $e^{iq \cdot r} \rightarrow 1$. Of the six blocks, only the pieces going like $\frac{1}{r^2}$ survive:

$$\frac{T(r)}{\xi D} \xrightarrow{r \rightarrow 0} \frac{T_{UV}(r)}{s^2}, \quad T_{UV}(r) = d_{\perp} \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2 r^2}.$$

(The UV pieces of T_3, T_5, T_6 cancel pointwise.) Since $\int d^2 r \frac{1}{r^2}$ diverges logarithmically at small r , the divergence is a log. **A log needs a second scale** to make its argument dimensionless, and that scale is set by how the subtraction term is cut off at large r :

- In the earlier PDF I subtracted $e^{iq \cdot r} \frac{T_{UV}}{s^2}$. Its large- r cut-off is the oscillation at $|r| \sim \frac{1}{q}$, so the log came out as $\log \frac{\mu^2}{q^2} = \log \frac{\mu^2}{\xi^2 k^2}$.
- To get $\log \mu^2(y-x)^2$ instead, subtract a term **without the phase**, cut off at $|r| \sim |s| = |y-x|$.

Both choices are legitimate. Each is just a way of writing 0 as $(-S + S)$; the physics is in the sum.

1. The coordinate-space counterterm

$$S(r) \equiv \frac{T_{UV}(r)}{r^2 + s^2} = \frac{T_{UV}(r)}{s^2} \cdot \frac{s^2}{r^2 + s^2}.$$

It does three jobs.

1. **Same UV behaviour.** $\frac{s^2}{r^2 + s^2} = 1 - \frac{r^2}{s^2} + \dots$, so $S = \frac{T_{UV}}{s^2} + O(r^0)$. The difference between the full integrand and S is then $O(1/r)$ (odd in $\hat{r}$) plus $O(r^0)$, which is integrable at $r = 0$.
2. **Convergent at large r without a phase.** $T_{UV} \sim \frac{1}{r^2}$, so $S \sim \frac{1}{r^4}$ and $\int d^2 r S$ converges at $|r| \rightarrow \infty$. The only scale left in it is s^2 .
3. **Trivial angular integral.** Apart from its numerator, S depends on r only through $|r|$ (step 3).

Any $\frac{T_{UV}}{r^2+a^2}$ would do. $a = |y - x|$ is chosen because it is the transverse scale already in the problem. A different a moves $\pi\mathcal{W}C_{UV} \log \frac{a^2}{s^2}$ between the UV term and the remainder, and the sum does not change.

2. Subtract and add

$$\int d^2r e^{iq \cdot r} \frac{T(r)}{\xi D} = \underbrace{\mu^{-2\epsilon} \int d^d r S(r)}_{\text{UV, done in } d=2-2\epsilon} + \underbrace{\int d^2r \left[e^{iq \cdot r} \frac{T(r)}{\xi D} - \frac{T_{UV}(r)}{r^2 + s^2} \right]_{d_\perp=2}}_{\equiv \tilde{\mathcal{Z}}_b(\xi), \text{ finite}}.$$

The factor $\mu^{-2\epsilon}$ restores the mass dimension, as before. The second integral is finite in $d = 2$, so $d_\perp \rightarrow 2$ there costs only $O(\epsilon)$.

Check that $\tilde{\mathcal{Z}}_b$ is UV finite. The angular average of the bracket on a small circle $|r| = \rho$, times ρ^2 , must vanish if no $\frac{1}{r^2}$ is left. It does, and it falls like ρ^2 :

ξ	$\rho = 10^{-2}$	$\rho = 10^{-3}$	$\rho = 10^{-4}$
0.3	$3.8 \cdot 10^{-4}$	$3.8 \cdot 10^{-6}$	$3.8 \cdot 10^{-8}$
0.7	$4.7 \cdot 10^{-4}$	$4.7 \cdot 10^{-6}$	$4.7 \cdot 10^{-8}$

3. Angular reduction of the counterterm (exact in d dimensions)

The rule. For any function $f(r^2)$,

$$\int d^d r r^i r^m f(r^2) = \frac{\delta^{im}}{d} \int d^d r r^2 f(r^2).$$

Why. The left side is a rank-2 tensor built from no external vector, so it must be $A \delta^{im}$. Taking the trace, $\delta^{ii} = d$, gives $A d = \int r^2 f$.

Apply it to S . Take $f = \frac{1}{r^2(r^2+s^2)}$ in the first term of T_{UV} :

$$\int d^d r S = \left[d_\perp \xi \bar{\xi} \frac{P^i s^m}{P^2} \frac{\delta^{im}}{d} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2} \right] \int \frac{d^d r}{r^2(r^2+s^2)} = \left[\frac{d_\perp}{d} \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right] \frac{P \cdot s}{P^2} \int \frac{d^d r}{r^2(r^2+s^2)}.$$

$d_\perp$ is the transverse dimension the vertices are contracted in. Here it is the same $d = 2 - 2\epsilon$ as the integral: that was the convention behind the $\frac{d_\perp}{2} M_0 = M_0 + \pi$ step of the earlier PDF. So $\frac{d_\perp}{d} = 1$ **exactly**, and

$$\int d^d r S = C_{UV}(\xi) \frac{P \cdot s}{P^2} \int \frac{d^d r}{r^2(r^2+s^2)} = C_{UV}(\xi) \mathcal{W} s^2 \int \frac{d^d r}{r^2(r^2+s^2)}.$$

Difference from the momentum-space version. There, the phase $e^{iq \cdot r}$ singled out the direction $\hat{q}$. The tensor integral then had a traceless part $-\pi \hat{q} \hat{q}$, and the $d_\perp$ trap left the finite piece $\pi \xi \bar{\xi} \mathcal{T}$. Here nothing picks a direction, so neither piece appears in the UV term. The $\pi \xi \bar{\xi} \mathcal{T}$ has not been lost: it now sits inside $\tilde{\mathcal{Z}}_b$ (step 7.1).

4. The scalar master integral, step by step

$$J_0 \equiv \mu^{-2\epsilon} \int \frac{d^d r}{r^2(r^2+s^2)}, \quad d = 2 - 2\epsilon.$$

(a) Radial form. Write $d^d r = r^{d-1} dr d\Omega_d$ with $\int d\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$. Put $u = r^2$, so $r^{d-1} dr = \frac{1}{2} u^{d/2-1} du$. For any f :

$$\int d^d r f(r^2) = \frac{\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty du u^{d/2-1} f(u).$$

(Check at $d = 2$: $\pi \int du f(u) = \int_0^\infty 2\pi r dr f(r^2)$.)

(b) Our integrand. $f(u) = \frac{1}{u(u+s^2)}$ gives $u^{d/2-1} f(u) = \frac{u^{a-1}}{u+c}$ with

$$a = \frac{d}{2} - 1 = -\epsilon, \quad c = s^2.$$

(c) **The Beta integral.** Substitute $u = ct$:

$$\int_0^\infty \frac{u^{a-1}}{u+c} du = c^{a-1} \int_0^\infty \frac{t^{a-1}}{1+t} dt = c^{a-1} B(a, 1-a) = c^{a-1} \Gamma(a) \Gamma(1-a).$$

This uses $B(a, b) = \int_0^\infty \frac{t^{a-1}}{(1+t)^{a+b}} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ with $b = 1-a$, $\Gamma(1) = 1$.

- It converges for $0 < a < 1$, i.e. $-1 < \epsilon < 0$.
- The small- u (UV) end is what needs $a > 0$, i.e. $\epsilon < 0$. This is why the pole is written $\frac{1}{\epsilon_{UV}} \equiv -\frac{1}{\epsilon} > 0$.
- The large- u end converges for any $a < 1$: the counterterm has no IR problem.

(d) **Put it together.** With $\frac{\pi^{d/2}}{\Gamma(d/2)} = \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}$:

$$J_0 = \mu^{-2\epsilon} \frac{\pi^{1-\epsilon}}{\Gamma(1-\epsilon)} \Gamma(-\epsilon) \Gamma(1+\epsilon) (s^2)^{-\epsilon-1} = \frac{\pi}{s^2} (\pi \mu^2 s^2)^{-\epsilon} \Gamma(-\epsilon) \frac{\Gamma(1+\epsilon)}{\Gamma(1-\epsilon)}.$$

(e) **Expand in ϵ .** Use $\Gamma(1+x) = 1 - \gamma_E x + O(x^2)$. Term by term:

$$\begin{aligned} \Gamma(-\epsilon) &= \frac{\Gamma(1-\epsilon)}{-\epsilon} = -\frac{1}{\epsilon} - \gamma_E + O(\epsilon), & \frac{\Gamma(1+\epsilon)}{\Gamma(1-\epsilon)} &= (1 - \gamma_E \epsilon)(1 + \gamma_E \epsilon) + O(\epsilon^2) = 1 - 2\gamma_E \epsilon + O(\epsilon^2), \\ (\pi \mu^2 s^2)^{-\epsilon} &= 1 - \epsilon \log(\pi \mu^2 s^2) + O(\epsilon^2). \end{aligned}$$

Multiply them out. Only the $\frac{1}{\epsilon}$ times the $O(\epsilon)$ terms give finite contributions:

$$\begin{aligned} \left(-\frac{1}{\epsilon} - \gamma_E\right)(1 - 2\gamma_E \epsilon)(1 - \epsilon \log \pi \mu^2 s^2) &= -\frac{1}{\epsilon} - \gamma_E + 2\gamma_E + \log(\pi \mu^2 s^2) + O(\epsilon). \\ \Rightarrow J_0 &= \frac{\pi}{s^2} \left[-\frac{1}{\epsilon} + \gamma_E + \log(\pi \mu^2 s^2)\right] + O(\epsilon). \end{aligned}$$

(f) $\overline{\text{MS}}$. By definition $\mu_{\overline{\text{MS}}}^2 = 4\pi \mu^2 e^{-\gamma_E}$, i.e. $\pi \mu^2 = \frac{\mu_{\overline{\text{MS}}}^2 e^{\gamma_E}}{4}$. So $\gamma_E + \log(\pi \mu^2 s^2) = \log \frac{\mu_{\overline{\text{MS}}}^2 s^2 e^{2\gamma_E}}{4}$, and:

$$J_0 = \frac{\pi}{s^2} \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 s^2 e^{2\gamma_E}}{4} \right], \quad s^2 = (y-x)^2.$$

Checks.

- sympy's series of the exact Γ -function expression gives $\log(\pi \mu^2 s^2) + \gamma_E - \frac{1}{\epsilon}$.
- The $\overline{\text{MS}}$ rewriting differs from it by 0.
- With a hard cut-off $|r| > \rho$ in $d = 2$, partial fractions $\frac{1}{u(u+c)} = \frac{1}{c} \left(\frac{1}{u} - \frac{1}{u+c} \right)$ give $\frac{\pi}{s^2} \log \frac{\rho^2 + s^2}{\rho^2} \rightarrow \frac{\pi}{s^2} \log \frac{s^2}{\rho^2}$. This is the same $\log s^2$, with $\frac{1}{\epsilon_{UV}} \leftrightarrow \log \frac{1}{\rho^2} + \text{const.}$

5. The UV term of part (b)

Steps 3 and 4 together: $\mu^{-2\epsilon} \int d^d r S = C_{UV} \mathcal{W} s^2 J_0$, i.e.

$$\mu^{-2\epsilon} \int d^d r \frac{T_{UV}(r)}{r^2 + s^2} = \pi \mathcal{W} C_{UV}(\xi) \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right].$$

Insert this into step 2. Pull one π into the prefactor, so that $\frac{g^4}{16\pi^5} \rightarrow \frac{g^4}{16\pi^4}$, i.e. $\mathcal{N}$:

$$\begin{aligned} \frac{d^3 N_b}{d^3 k} &= \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \int_\lambda^{1-\lambda} d\xi \left\{ \mathcal{W} C_{UV}(\xi) \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right] + \frac{1}{\pi} \tilde{\mathcal{L}}_b(\xi) \right\} \mathcal{U}_b, \\ \tilde{\mathcal{L}}_b(\xi) &= \int d^2 r \left[e^{i\xi k \cdot r} \frac{T(r)}{\xi D} - \frac{T_{UV}(r)}{r^2 + s^2} \right]_{d=2}. \end{aligned}$$

This is the full z integral of part (b). Nothing has been dropped.

6. The p^+ integral of the UV term

The bracket $[\frac{1}{\epsilon_{UV}} + \log \dots]$ does **not depend on** ξ . Before, it contained $\log \frac{1}{\xi^2}$ through $q = \bar{\xi}k$; now it does not. So only the plain moment of C_{UV} is needed:

$$\int_{\lambda}^{1-\lambda} d\xi C_{UV}(\xi) = \underbrace{\frac{1}{6}}_{\xi\bar{\xi}} + \underbrace{(L-1)}_{\xi/\bar{\xi}} + \underbrace{(L-1)}_{\bar{\xi}/\xi} = 2L - \frac{11}{6} + O(\lambda).$$

Here $\int_{\lambda}^{1-\lambda} \frac{\xi}{\bar{\xi}} d\xi = \int_{\lambda}^{1-\lambda} (\frac{1}{\xi} - 1) d\xi = \log \frac{1-\lambda}{\lambda} - (1-2\lambda) \rightarrow L-1$, and the same for $\frac{\bar{\xi}}{\xi}$. The UV term of part (b) is therefore

$$\mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \mathcal{W} \left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{\overline{\text{MS}}}^2 (y-x)^2 e^{2\gamma_E}}{4} \right] \mathcal{U}_b.$$

It contains no L^2 and no $\frac{\pi^2}{3} - \frac{67}{18}$: those came from the $\log \frac{1}{\xi^2}$, which is gone. They reappear in the remainder (§7).

7. The remainder $\tilde{\mathcal{I}}_b$

7.1 Exact relation to the momentum-space remainder

The earlier remainder is $\mathcal{I}_b = \int d^2r e^{iq \cdot r} [\frac{T}{\xi D} - \frac{T_{UV}}{s^2}]$. The two remainders differ only in the subtraction term:

$$\tilde{\mathcal{I}}_b - \mathcal{I}_b = \int d^2r T_{UV}(r) \left[\frac{e^{iq \cdot r}}{s^2} - \frac{1}{r^2 + s^2} \right].$$

This is finite: at small r the bracket is $O(r)$, and at large r the first term oscillates while the second falls like $\frac{1}{r^4}$. So it can be done directly in $d = 2$. Use polar coordinates with $\hat{q}$ along the first axis, $\hat{r} = (\cos \phi, \sin \phi)$, and $x = \frac{|r|}{|s|}$.

Angular integrals (Jacobi–Anger: $\int_0^{2\pi} d\phi e^{ia \cos \phi} \cos n\phi = 2\pi i^n J_n(a)$; the $\sin n\phi$ terms vanish by $\phi \rightarrow -\phi$):

- Scalar: $\int d\phi e^{iq \cdot r} = 2\pi J_0(qr)$.
- Tensor: $\hat{r}^i \hat{r}^m = \frac{1}{2} \delta^{im} + \frac{1}{2} \cos 2\phi (2\hat{q}^i \hat{q}^m - \delta^{im}) + \frac{1}{2} \sin 2\phi (\dots)$. So $\int d\phi e^{iq \cdot r} \hat{r}^i \hat{r}^m = \pi \delta^{im} J_0(qr) - \pi (2\hat{q}^i \hat{q}^m - \delta^{im}) J_2(qr)$.
- Without the phase: $\int d\phi \hat{r}^i \hat{r}^m = \pi \delta^{im}$.

Radial integrals.

$$(R1) \quad \int_0^\infty \frac{dx}{x} \left[J_0(ax) - \frac{1}{1+x^2} \right] = -\frac{1}{2} \log \frac{a^2 e^{2\gamma_E}}{4}, \quad (R2) \quad \int_0^\infty \frac{dx}{x} J_2(ax) = \frac{1}{2}.$$

Derivation of (R1). Insert x^η and let $\eta \rightarrow 0^+$ at the end. The two Mellin transforms are

$$\int_0^\infty x^{\eta-1} J_0(ax) dx = a^{-\eta} 2^{\eta-1} \frac{\Gamma(\eta/2)}{\Gamma(1-\eta/2)}, \quad \int_0^\infty \frac{x^{\eta-1}}{1+x^2} dx = \frac{1}{2} \Gamma\left(\frac{\eta}{2}\right) \Gamma\left(1 - \frac{\eta}{2}\right).$$

With $\Gamma(\frac{\eta}{2}) = \frac{2}{\eta} - \gamma_E + O(\eta)$ and $\Gamma(1 \mp \frac{\eta}{2}) = 1 \pm \frac{\gamma_E \eta}{2}$:

- the first transform is $\frac{1}{\eta} - \gamma_E - \log \frac{a}{2} + O(\eta)$;
- the second is $\frac{1}{\eta} + O(\eta)$.

Their difference is $-\gamma_E - \log \frac{a}{2} = -\frac{1}{2} \log \frac{a^2 e^{2\gamma_E}}{4}$.

(R2) is the standard $\int_0^\infty J_\nu(t) \frac{dt}{t} = \frac{1}{\nu}$. It needs no subtraction, because $J_2 \sim \frac{x^2}{8}$ at small x .

Numerical checks (src/rowV_coordUV.py, §4):

$a = q s $	(R1) by quadrature	$-\frac{1}{2} \log \frac{a^2 e^{2\gamma_E}}{4}$
0.3	1.31990431664	1.31990431998
1.0	0.115931518961	0.115931515658
3.0	-0.982680744912	-0.98268077301

(R2) by quadrature gives 0.5.

Assemble. Put $d^2 r = s^2 x dx d\phi$.

Scalar part, $(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}) \frac{P \cdot s}{P^2 s^2}$:

$$\left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}\right) \frac{P \cdot s}{P^2 s^2} 2\pi \int_0^\infty \frac{dx}{x} \left[J_0(q|x) - \frac{1}{1+x^2} \right] = -\pi \mathcal{W} \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \log \frac{q^2 s^2 e^{2\gamma_E}}{4}.$$

Tensor part, $2\xi\bar{\xi} \frac{P^i s^m}{P^2} \frac{\hat{r}^i \hat{r}^m}{r^2}$ at $d_\perp = 2$:

$$\frac{2\xi\bar{\xi}}{P^2 s^2} \int_0^\infty \frac{dx}{x} \left\{ \pi(P \cdot s) \left[J_0 - \frac{1}{1+x^2} \right] - \pi [2(P \cdot \hat{q})(s \cdot \hat{q}) - P \cdot s] J_2 \right\} = -\pi \xi \bar{\xi} \mathcal{W} \log \frac{q^2 s^2 e^{2\gamma_E}}{4} + \pi \xi \bar{\xi} \mathcal{T}.$$

In the last step $\hat{q} = \hat{k}$, because $q = \bar{\xi} k$ with $\bar{\xi} > 0$. Adding the two parts, and writing $\log q^2 = \log \bar{\xi}^2 + \log k^2$:

$$\tilde{\mathcal{I}}_b(\xi) = \mathcal{I}_b(\xi) + \pi \xi \bar{\xi} \mathcal{T} - \pi \mathcal{W} C_{UV}(\xi) [\log \bar{\xi}^2 + \ell], \quad \ell \equiv \log \frac{k^2 (y-x)^2 e^{2\gamma_E}}{4}.$$

Independent cross-check. Take the difference of the two dimensionally regulated counterterms: $(\pi \mathcal{W} C_{UV}[\frac{1}{\epsilon_{UV}} + \log \frac{\mu^2}{q^2}] + \pi \xi \bar{\xi} \mathcal{T}) - \pi \mathcal{W} C_{UV}[\frac{1}{\epsilon_{UV}} + \log \frac{\mu^2 s^2 e^{2\gamma_E}}{4}]$. The pole and μ cancel, and the result is exactly the line above. So the two counterterms have the same $\frac{1}{\epsilon_{UV}}$ coefficient, as they must.

7.2 Behaviour at the endpoints in ξ

From the earlier PDF (§6.2, measured by 2D quadrature):

$$\mathcal{I}_b \xrightarrow{\xi \rightarrow 0} \frac{\beta_0}{\xi}, \quad \mathcal{I}_b \xrightarrow{\xi \rightarrow 1} \frac{-\pi \mathcal{W} \log \frac{1}{\xi} + \beta_1}{\bar{\xi}}.$$

The added terms, at each end:

- $\xi \rightarrow 0$: $C_{UV} \rightarrow \frac{1}{\xi}$, $\log \bar{\xi}^2 \rightarrow 0$, $\xi \bar{\xi} \mathcal{T} \rightarrow 0$. So they add $-\frac{\pi \mathcal{W} \ell}{\xi}$.
- $\xi \rightarrow 1$: $C_{UV} \rightarrow \frac{1}{\bar{\xi}}$, $\log \bar{\xi}^2 = -2 \log \frac{1}{\bar{\xi}}$. So they add $\frac{2\pi \mathcal{W} \log \frac{1}{\bar{\xi}} - \pi \mathcal{W} \ell}{\bar{\xi}}$.

Hence

$$\tilde{\mathcal{I}}_b \xrightarrow{\xi \rightarrow 0} \frac{\tilde{\beta}_0}{\xi}, \quad \tilde{\mathcal{I}}_b \xrightarrow{\xi \rightarrow 1} \frac{+\pi \mathcal{W} \log \frac{1}{\bar{\xi}} + \tilde{\beta}_1}{\bar{\xi}}, \quad \tilde{\beta}_0 = \beta_0 - \pi \mathcal{W} \ell, \quad \tilde{\beta}_1 = \beta_1 - \pi \mathcal{W} \ell.$$

The sign of the $\log \frac{1}{\bar{\xi}}$ flips, and this is expected. As §6.3 of the earlier PDF showed, the true integrand carries $+\frac{\pi \mathcal{W}}{\xi} \log \frac{1}{\bar{\xi}}$, generated at large $|r|$ (between $|s|$ and $\frac{1}{q}$). The old counterterm ran out to $\frac{1}{q}$ and carried twice that. The new one stops at $|s|$ and carries none. So $\tilde{\mathcal{I}}_b$ keeps exactly the log of the true integrand.

7.3 The plus prescription

At $\xi \rightarrow 0$ it is a simple pole with a finite residue $\tilde{\beta}_0$, so the ordinary plus prescription works. **At $\xi \rightarrow 1$** there is a $\frac{\log}{\bar{\xi}}$, so two distributions are needed. Split

$$\tilde{\mathcal{I}}_b = \underbrace{\left[\tilde{\mathcal{I}}_b - \frac{\tilde{\beta}_0}{\xi} - \frac{\tilde{\beta}_1 + \pi \mathcal{W} \log \frac{1}{\bar{\xi}}}{\bar{\xi}} \right]}_{\text{integrable on } [0,1]} + \frac{\tilde{\beta}_0}{\xi} + \frac{\tilde{\beta}_1}{\bar{\xi}} + \pi \mathcal{W} \frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}}.$$

Integrate each piece over $[\lambda, 1 - \lambda]$:

- $\int_{\lambda}^{1-\lambda} \frac{d\xi}{\xi} = \log \frac{1-\lambda}{\lambda} = L + O(\lambda)$, and the same for $\frac{1}{\xi}$.
- With $v = \bar{\xi}$: $\int_{\lambda}^{1-\lambda} \frac{\log(1/v)}{v} dv = \left[-\frac{1}{2} \log^2 v \right]_{\lambda}^{1-\lambda} = \frac{1}{2} \log^2 \lambda - \frac{1}{2} \log^2(1 - \lambda) = \frac{1}{2} L^2 + O(\lambda^2)$.
- The bracket is integrable at both ends, so its limits can go to 0 and 1 at a cost of $O(\lambda)$.

In distribution language this is $\frac{1}{\xi} = [\frac{1}{\xi}]_+ + \delta(\xi)L$, $\frac{1}{\bar{\xi}} = [\frac{1}{\bar{\xi}}]_+ + \delta(\bar{\xi})L$, $\frac{\log(1/\bar{\xi})}{\bar{\xi}} = [\frac{\log(1/\bar{\xi})}{\bar{\xi}}]_+ + \frac{1}{2} \delta(\bar{\xi})L^2$. Therefore

$$\int_{\lambda}^{1-\lambda} d\xi \tilde{\mathcal{I}}_b(\xi) = (\tilde{\beta}_0 + \tilde{\beta}_1)L + \frac{\pi\mathcal{W}}{2}L^2 + \tilde{\mathcal{F}}_b + O(\Lambda),$$

$$\tilde{\mathcal{F}}_b = \int_0^1 d\xi \left[\tilde{\mathcal{I}}_b(\xi) - \frac{\tilde{\beta}_0}{\xi} - \frac{\tilde{\beta}_1 + \pi\mathcal{W} \log \frac{1}{\xi}}{\bar{\xi}} \right], \quad \tilde{\beta}_0 = \lim_{\xi \rightarrow 0} \xi \tilde{\mathcal{I}}_b, \quad \tilde{\beta}_1 = \lim_{\xi \rightarrow 1} \left[\bar{\xi} \tilde{\mathcal{I}}_b - \pi\mathcal{W} \log \frac{1}{\bar{\xi}} \right].$$

$\tilde{\mathcal{F}}_b$ in terms of the old $\mathcal{F}_b$. Subtract the two definitions and use 7.1, with the moments

- $\int_0^1 (C_{UV} - \frac{1}{\xi} - \frac{1}{\bar{\xi}}) = \int_0^1 (\xi\bar{\xi} - 2) = \frac{1}{6} - 2$;
- $\int_0^1 (C_{UV} - \frac{1}{\xi}) \log \frac{1}{\xi^2} = \frac{5}{18} - 2 + \frac{\pi^2}{3} - 2$ (table in §6.1 of the earlier PDF).

$$\tilde{\mathcal{F}}_b = \mathcal{F}_b + \pi \left[\mathcal{W} \left(\frac{\pi^2}{3} - \frac{67}{18} + \frac{11}{6} \ell \right) + \frac{1}{6} \mathcal{T} \right].$$

8. Final result for part (b), and the consistency check

Add §6 and $\frac{1}{\pi} \times (7.3)$. This gives the formula in the box on page 1:

$$\frac{d^3 N_b}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \left\{ \mathcal{W} \left(2L - \frac{11}{6} \right) \left[\frac{1}{\epsilon_{UV}} + \log \frac{\mu_{MS}^2 (y - x)^2 e^{2\gamma_E}}{4} \right] + \frac{1}{2} \mathcal{W} L^2 + \frac{\tilde{\beta}_0 + \tilde{\beta}_1}{\pi} L + \frac{\tilde{\mathcal{F}}_b}{\pi} \right\} \mathcal{U}_b$$

Consistency with the momentum-space form. The momentum-space answer was

$$\mathcal{W} \left[\left(2L - \frac{11}{6} \right) \left(\frac{1}{\epsilon_{UV}} + \log \frac{\mu^2}{k^2} \right) + \frac{1}{2} L^2 + \frac{\pi^2}{3} - \frac{67}{18} \right] + \frac{1}{6} \mathcal{T} + \frac{\beta_0 + \beta_1}{\pi} L + \frac{\mathcal{F}_b}{\pi}.$$

Subtract it from the bracket above, term by term, using $\log \frac{\mu^2 s^2 e^{2\gamma_E}}{4} = \log \frac{\mu^2}{k^2} + \ell$:

source	coordinate – momentum
UV term	$+\mathcal{W} \left(2L - \frac{11}{6} \right) \ell - \mathcal{W} \left(\frac{\pi^2}{3} - \frac{67}{18} \right) - \frac{1}{6} \mathcal{T}$
L^2	$\frac{1}{2} \mathcal{W} L^2 - \frac{1}{2} \mathcal{W} L^2 = 0$
single L	$\frac{(\tilde{\beta}_0 - \beta_0) + (\tilde{\beta}_1 - \beta_1)}{\pi} L = -2\mathcal{W} \ell L$
finite	$\frac{\tilde{\mathcal{F}}_b - \mathcal{F}_b}{\pi} = \mathcal{W} \left(\frac{\pi^2}{3} - \frac{67}{18} \right) + \frac{11}{6} \mathcal{W} \ell + \frac{1}{6} \mathcal{T}$
sum	0

So both forms are the same result. What moved:

- The $\frac{1}{2} \mathcal{W} L^2$ now comes entirely from the remainder. Before, it was L^2 from the UV term minus $\frac{1}{2} L^2$ from the remainder.
- The constants $\frac{\pi^2}{3} - \frac{67}{18}$ and $\frac{1}{6} \mathcal{T}$ are now inside $\tilde{\mathcal{F}}_b$.

What was checked numerically, and what follows by algebra.

- **Checked numerically:** J_0 (series and $\overline{\text{MS}}$ form); the isotropy of the counterterm; the UV finiteness of $\tilde{\mathcal{I}}_b$ (ρ^2 fall-off); the radial integrals (R1), (R2).
- **By algebra:** the endpoint coefficients of $\tilde{\mathcal{I}}_b$ follow exactly from 7.1 and the previously measured endpoints of $\mathcal{I}_b$ (slopes 3.16, 3.11, 3.07, 3.06, 3.09 against $-\pi\mathcal{W} = 3.061$).
- **Caveat:** the soft- x caveat of the earlier PDF (§9) is unchanged. It is the non-oscillating $\frac{1}{(y-x)^2}$ tail of β_0 . The new piece $-\pi\mathcal{W}\ell$ in $\tilde{\beta}_0$ keeps the phase $e^{ik \cdot x}$ and is not of that type.

Checks: `src/rowV_coordUV.py` (§1 scalar master, §2 isotropy, §3 UV finiteness of the remainder, §4 the Bessel relation to the momentum-space subtraction, §5 the ξ endpoints), output in `RESULTS_rowV_coordUV.txt`. Earlier inputs: `src/rowV_final.py`, `src/row28_rem.py`.